

```
!pip install -q pyspark
```

```
from pyspark.sql import SparkSession
```

```
spark = SparkSession.builder \
    .master("local[*]") \
    .appName("ColabSpark") \
    .getOrCreate()

print("Spark Version:", spark.version)
```

Spark Version: 4.0.2

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```
df = spark.read.csv("/content/drive/MyDrive/Datasets/heroes_information.csv", header = True)
```

```
df.show()
```

Id	Name	Gender	Eyecolor	Race	Haircolor	Height	Publisher	Skincolor	Alignment	Weight
0	A-Bomb	Male	yellow	Human	No Hair	203	Marvel Comics	-	good	441
1	Abe Sapien	Male	blue	Ichthy Sapien	No Hair	191	Dark Horse Comics	blue	good	65
2	Abin Sur	Male	blue	Ungaran	No Hair	185	DC Comics	red	good	90
3	Abomination	Male	green	Human / Radiation	No Hair	203	Marvel Comics	-	bad	441
4	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel Comics	-	bad	-99
5	Absorbing Man	Male	blue	Human	No Hair	193	Marvel Comics	-	bad	122
6	Adam Monroe	Male	blue	-	Blond	-99	NBC - Heroes	-	good	-99
7	Adam Strange	Male	blue	Human	Blond	185	DC Comics	-	good	88
8	Agent 13	Female	blue	-	Blond	173	Marvel Comics	-	good	61
9	Agent Bob	Male	brown	Human	Brown	178	Marvel Comics	-	good	81
10	Agent Zero	Male	-	-	-	191	Marvel Comics	-	good	104
11	Air-Walker	Male	blue	-	White	188	Marvel Comics	-	bad	108
12	Ajax	Male	brown	Cyborg	Black	193	Marvel Comics	-	bad	90
13	Alan Scott	Male	blue	-	Blond	180	DC Comics	-	good	90
14	Alex Mercer	Male	-	Human	-	-99	Wildstorm	-	bad	-99
15	Alex Woosly	Male	-	-	-	-99	NBC - Heroes	-	good	-99
16	Alfred Pennyworth	Male	blue	Human	Black	178	DC Comics	-	good	72
17	Alien	Male	-	Xenomorph XX121	No Hair	244	Dark Horse Comics	black	bad	169
18	Allan Quatermain	Male	-	-	-	-99	Wildstorm	-	good	-99
19	Amazo	Male	red	Android	-	257	DC Comics	-	bad	173

only showing top 20 rows

```
df.show(10)
```

Id	Name	Gender	Eyecolor	Race	Haircolor	Height	Publisher	Skincolor	Alignment	Weight
0	A-Bomb	Male	yellow	Human	No Hair	203	Marvel Comics	-	good	441
1	Abe Sapien	Male	blue	Ichthy Sapien	No Hair	191	Dark Horse Comics	blue	good	65
2	Abin Sur	Male	blue	Ungaran	No Hair	185	DC Comics	red	good	90
3	Abomination	Male	green	Human / Radiation	No Hair	203	Marvel Comics	-	bad	441
4	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel Comics	-	bad	-99
5	Absorbing Man	Male	blue	Human	No Hair	193	Marvel Comics	-	bad	122
6	Adam Monroe	Male	blue	-	Blond	-99	NBC - Heroes	-	good	-99
7	Adam Strange	Male	blue	Human	Blond	185	DC Comics	-	good	88
8	Agent 13	Female	blue	-	Blond	173	Marvel Comics	-	good	61
9	Agent Bob	Male	brown	Human	Brown	178	Marvel Comics	-	good	81

only showing top 10 rows

```
male_heroes = df.filter(df['Gender'] == 'Male').count()
male_heroes
```

505

```
female_heroes = df.filter(df['Gender'] == 'Female').count()
female_heroes
```

200

```
race = df.filter(df['Race'] == 'Android').count()
race
```

9

```
from pyspark.sql.functions import col

df_cleaned = df.filter(col('Height').cast('float') > 0).withColumn('Height', col('Height').cast('float'))

average_height_by_publisher = df_cleaned.groupBy('Publisher').agg({'Height': 'avg'})

average_height_by_publisher = average_height_by_publisher.withColumnRenamed('avg(Height)', 'Average Height')

average_height_by_publisher.orderBy(col('Average Height').desc()).show(truncate=False)
```

Publisher	Average Height
J. R. R. Tolkien	279.0
Microsoft	213.0
Image Comics	211.0
Marvel Comics	190.51080243087108
Rebellion	188.0
George Lucas	183.91666666666666
Titan Books	183.0
Team Epic TV	182.2
Star Trek	181.5
DC Comics	180.90068493150685
Sony Pictures	179.0
Dark Horse Comics	176.9090909090909
Shueisha	171.5
NULL	142.58333333333334

```
df.createOrReplaceTempView("superheroes")

blue_eyes_sql = spark.sql("SELECT * FROM superheroes WHERE Eyecolor = 'blue'")
blue_eyes_sql.show()
```

Id	Name	Gender	Eyecolor	Race	Haircolor	Height	Publisher	Skincolor	Alignment	Weight
1	Abe Sapien	Male	blue	Ichthyo Sapien	No Hair	191	Dark Horse Comics	blue	good	65
2	Abin Sur	Male	blue	Ungaran	No Hair	185	DC Comics	red	good	90
4	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel Comics	-	bad	-99
5	Absorbing Man	Male	blue	Human	No Hair	193	Marvel Comics	-	bad	122
6	Adam Monroe	Male	blue	-	Blond	-99	NBC - Heroes	-	good	-99
7	Adam Strange	Male	blue	Human	Blond	185	DC Comics	-	good	88
8	Agent 13	Female	blue	-	Blond	173	Marvel Comics	-	good	61
11	Air-Walker	Male	blue	-	White	188	Marvel Comics	-	bad	108
13	Alan Scott	Male	blue	-	Blond	180	DC Comics	-	good	90
16	Alfred Pennyworth	Male	blue	Human	Black	178	DC Comics	-	good	72
22	Angel	Male	blue	-	Blond	183	Marvel Comics	-	good	68
27	Animal Man	Male	blue	Human	Blond	183	DC Comics	-	good	83
29	Ant-Man	Male	blue	Human	Blond	211	Marvel Comics	-	good	122
30	Ant-Man II	Male	blue	Human	Blond	183	Marvel Comics	-	good	86
33	Anti-Venom	Male	blue	Symbiote	Blond	229	Marvel Comics	-	-	358
35	Aquababy	Male	blue	-	Blond	-99	DC Comics	-	good	-99
36	Aqualad	Male	blue	Atlantean	Black	178	DC Comics	-	good	106
37	Aquaman	Male	blue	Atlantean	Blond	185	DC Comics	-	good	146
38	Arachne	Female	blue	Human	Blond	175	Marvel Comics	-	good	63
39	Archangel	Male	blue	Mutant	Blond	183	Marvel Comics	blue	good	68

only showing top 20 rows